MICROSOFT AZURE

Complete Textbook

Cloud Computing Fundamentals to Advanced Architecture

✦ 2025 Edition ✦

Covers Exam Objectives

AZ-900 • AZ-104 • AZ-204 • AZ-305 • AZ-500 • AZ-400

60+ Regions • 200+ Services • 99.99% SLA

Microsoft Azure | 2025 Edition

Chapter 1: Introduction to Cloud Computing & Microsoft Azure

1.1 What is Cloud Computing?

Cloud computing is the delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the Internet ("the cloud") to offer faster innovation, flexible resources, and economies of scale. Organizations pay only for the cloud services they use, which helps lower operating costs, run infrastructure more efficiently, and scale as business needs change.

Before cloud computing, organizations maintained costly on-premises infrastructure requiring significant capital expenditure, dedicated IT staff, and long provisioning lead times. Cloud computing fundamentally transformed this model, moving organizations from a CapEx (capital expenditure) model to an OpEx (operational expenditure) model.

> **Key Definition**
> Cloud Computing: On-demand availability of computer system resources—especially data storage and computing power—without direct active management by the user.
>
> Pay-as-you-go: You only pay for the services you consume, eliminating the need for large upfront capital expenditure. Resources can be provisioned in minutes and released when no longer needed.

1.2 Cloud Service Models

Cloud computing is offered in three primary service models, each providing different levels of abstraction and management responsibility. Understanding these models is fundamental to selecting the right solution for each workload.

Model	Full Name	What You Manage	What Provider Manages	Azure Example
IaaS	Infrastructure as a Service	OS, Middleware, Runtime, Apps, Data	Virtualization, Servers, Storage, Networking	Azure VMs, Azure Disk Storage
PaaS	Platform as a Service	Applications and Data only	Infrastructure + OS + Runtime + Middleware	Azure App Service, Azure SQL DB
SaaS	Software as a Service	Configuration and user data	Everything including the application	Microsoft 365, Dynamics 365
FaaS	Function as a Service (Serverless)	Application code logic only	All infrastructure and scaling	Azure Functions, Logic Apps

The Shared Responsibility Model

The shared responsibility model defines who is responsible for security and compliance at each layer. As you move from IaaS to PaaS to SaaS, the provider takes on more responsibility and the customer's burden decreases.

- IaaS: Customer is responsible for OS, middleware, runtime, applications, and data. Provider handles physical infrastructure.
- PaaS: Customer manages applications and data. Provider handles OS, runtime, and underlying infrastructure.
- SaaS: Provider manages nearly everything. Customer is responsible only for user access, data classification, and endpoint security.

1.3 Cloud Deployment Models

Organizations can deploy cloud solutions using different architectural models depending on their regulatory requirements, existing investments, and strategic goals.

Model	Infrastructure	Control Level	Cost Profile	Best Use Case
Public Cloud	Shared, multi-tenant infrastructure	Low – Provider-managed	Pay-per-use, no CapEx	Startups, web apps, dev/test environments
Private Cloud	Dedicated, single-tenant infrastructure	High – Customer-managed	Higher cost, capital investment	Government, healthcare, finance (regulated industries)
Hybrid Cloud	Mix of public + private cloud	Medium – Split responsibility	Flexible, optimize for workload	Gradual migration, bursting, compliance-sensitive apps
Multi-Cloud	Multiple cloud providers simultaneously	Variable – Per provider	Variable, potentially complex billing	Vendor independence, best-of-breed services

1.4 Benefits of Cloud Computing

Microsoft Azure and cloud computing in general deliver transformative business benefits that allow organizations to innovate faster, operate more efficiently, and reduce costs.

- High Availability & Reliability: Azure SLAs guarantee uptime—99.99% for many services—backed by geo-redundant datacenters and automatic failover.
- Scalability: Instantly scale out (add VM instances) or scale up (increase instance size) to handle spikes in demand without pre-provisioning.
- Elasticity: Azure automatically adds or removes resources based on real-time demand signals, ensuring optimal cost and performance.
- Agility: Deploy fully configured infrastructure in minutes using ARM templates or Bicep, rather than weeks of procurement and configuration.
- Geo-Distribution: Deploy applications to 60+ Azure regions globally, reducing latency for end-users and enabling compliance with data sovereignty requirements.
- Disaster Recovery: Built-in redundancy, Azure Site Recovery, and geo-replication ensure business continuity even during regional outages.
- Cost Savings: Convert capital expenditure (CapEx) to operational expenditure (OpEx). Pay-as-you-go eliminates idle infrastructure costs.
- Security: Microsoft invests $1 billion+ annually in cybersecurity R&D and employs 3,500+ dedicated security professionals globally.
- Compliance: Azure meets 90+ compliance certifications including ISO 27001, SOC 2, HIPAA, FedRAMP, and GDPR requirements.

- Managed Services: Azure handles patching, upgrades, and operational tasks for PaaS services, freeing teams to focus on business value.

1.5 Microsoft Azure Overview

Microsoft Azure, commonly referred to as Azure, is Microsoft's public cloud computing platform—one of the largest globally, competing with AWS and Google Cloud. Azure was launched in February 2010 and has since grown to 60+ regions, 200+ services, and serves millions of customers worldwide, from startups to Fortune 500 enterprises.

Azure provides a comprehensive set of cloud services spanning compute, storage, networking, databases, AI/ML, analytics, IoT, security, and developer tools. It supports diverse programming languages, frameworks, and operating systems—including Windows and Linux.

Azure Infrastructure Term	Definition	Key Fact
Geography	A market (e.g., United States, Europe) containing 2+ Azure regions ensuring data residency compliance	Meets local regulatory and data sovereignty requirements
Region	A set of datacenters deployed within a latency-defined perimeter, connected via dedicated low-latency network	Azure has 60+ regions across 140+ countries
Region Pair	Each region is paired with another region in the same geography, at least 300 miles apart, for disaster recovery	Paired regions receive updates in sequence to prevent simultaneous outages
Availability Zone	Physically separate datacenters within a region with independent power, cooling, and networking	Minimum 3 zones per region; 1ms or less latency between zones
Availability Set	Logical grouping ensuring VMs are distributed across fault domains and update domains within a datacenter	Protects against hardware failures and planned maintenance events
Edge Zone	Extended Azure compute deployed at carrier locations or customer premises for ultra-low latency	Used for latency-sensitive workloads like gaming and video streaming

Chapter 2: Azure Core Services – Compute

Azure Compute is the foundation of cloud workloads, encompassing virtual machines, containers, serverless functions, and managed application hosting. Understanding compute options allows architects to select the right service for each workload type, balancing control, management overhead, scalability, and cost.

2.1 Azure Virtual Machines (VMs)

Azure Virtual Machines provide on-demand, scalable computing resources that give you the flexibility of virtualization without buying or maintaining physical hardware. VMs are ideal for lift-and-shift migrations, legacy application hosting, custom OS configurations, and workloads requiring full operating system control.

Each VM runs in an Azure region inside a Virtual Network (VNet). It has one or more disks (OS disk mandatory, data disks optional), a Network Interface Card (NIC), and associated compute, storage, and network costs.

VM Series	Name	vCPUs	RAM Range	Best Workloads
B-Series	Burstable General Purpose	1–20	0.5–80 GB	Dev/Test, low-traffic web, microservices with variable CPU
D-Series	General Purpose	2–96	8–384 GB	Enterprise applications, relational databases, web servers
E-Series	Memory Optimized	2–96	16–672 GB	In-memory databases (SAP HANA), large caches, analytics
F-Series	Compute Optimized	2–72	4–144 GB	Batch processing, gaming servers, scientific modeling
H-Series	High Performance Compute	8–80	32 GB–2 TB	Fluid dynamics, weather modeling, molecular simulations
N-Series	GPU Optimized	6–64	56–440 GB	AI/ML training, deep learning, rendering, CUDA workloads
L-Series	Storage Optimized	8–80	64–640 GB	NoSQL databases, big data, data warehousing
M-Series	Memory + Compute	8–416	220 GB–11.4 TB	Very large in-memory databases, SAP HANA scale-up

VM Disk Types

Disk Type	Max IOPS	Max Throughput	Best For	Redundancy
Ultra Disk	160,000 IOPS	2,000 MB/s	SAP HANA, top-tier SQL, mission-critical	LRS only
Premium SSD v2	80,000 IOPS	1,200 MB/s	Databases needing consistent sub-ms latency	LRS only
Premium SSD	20,000 IOPS	900 MB/s	Production databases, business-critical apps	LRS/ZRS
Standard SSD	6,000 IOPS	750 MB/s	Lightly used web servers, dev/test DBs	LRS/ZRS

Disk Type	Max IOPS	Max Throughput	Best For	Redundancy
Standard HDD	2,000 IOPS	500 MB/s	Backup, archival, non-critical workloads	LRS/ZRS/GRS

2.2 VM Scale Sets & Availability

Azure Virtual Machine Scale Sets (VMSS) let you create and manage a group of load-balanced, identical VMs. The number of VM instances can automatically increase or decrease in response to demand, a schedule, or custom metrics—making VMSS ideal for auto-scaling stateless workloads.

Concept	Description	Key Use Case
Availability Set	Distributes VMs across fault domains (separate hardware racks) and update domains (patching groups) within a datacenter	Protects against hardware failures and planned maintenance
Availability Zone	Deploys VMs across 3 physically separate zones within a region, each with independent power/cooling/networking	Zone-level protection; SLA 99.99% for VMs across zones
VM Scale Set	Automatically manages and scales a set of identical VM instances based on metric rules, schedules, or manual triggers	Web tier autoscaling, stateless microservices, batch compute
Spot VMs	Leverages Azure's unused capacity at up to 90% discount; can be evicted on 30-second notice when Azure needs capacity	Batch jobs, rendering, testing, fault-tolerant workloads
Reserved Instances	1 or 3-year reservation for specific VM size/region provides 40–72% savings over pay-as-you-go pricing	Stable production workloads with predictable demand

2.3 Azure App Service

Azure App Service is a fully managed HTTP-based Platform as a Service (PaaS) for hosting web applications, REST APIs, and mobile backends. It abstracts away infrastructure management, OS patching, and runtime upgrades—allowing developers to focus entirely on application code.

App Service supports multiple runtimes including .NET, Java, Node.js, Python, PHP, and Ruby. It features built-in CI/CD integration, custom domains, SSL certificates, autoscaling, and deployment slots for zero-downtime deployments.

App Service Plan Tier	Purpose	Key Features	SLA
Free / Shared (F1, D1)	Learning and experimentation only	Shared infrastructure, limited CPU minutes, no custom domain, no SLA	No SLA
Basic (B1, B2, B3)	Dev/test with dedicated compute	Dedicated VMs, manual scale (up to 3 instances), custom domains, SSL	No SLA

App Service Plan Tier	Purpose	Key Features	SLA
Standard (S1, S2, S3)	Production web apps	Auto-scale (up to 10 instances), deployment slots (5), custom SSL, SLA	99.95%
Premium v3 (P1v3–P3v3)	Performance-sensitive production apps	Faster CPU/RAM, VNet integration, 30 slots, private endpoints	99.95%
Isolated v2 (I1v2–I3v2)	Fully isolated, compliance-sensitive workloads	App Service Environment (ASE), private VNet injection, dedicated infrastructure	99.95%

2.4 Azure Kubernetes Service (AKS)

Azure Kubernetes Service (AKS) simplifies deploying, managing, and scaling containerized applications using Kubernetes. Azure handles critical tasks such as health monitoring, automatic node upgrades, and control plane management—you only pay for the worker nodes.

AKS is the preferred platform for microservices architectures, cloud-native applications, and teams practicing DevOps and GitOps workflows. It integrates natively with Azure Container Registry, Azure Monitor, Azure Policy, and Azure Active Directory.

AKS Component	Description	Management
Control Plane	kube-apiserver, etcd, kube-scheduler, kube-controller-manager	Fully managed by Azure (free of charge)
Node Pools	Groups of VM nodes running containerized workloads (pods)	Customer-managed; Azure handles node OS patching
Pods	Smallest deployable unit; one or more containers sharing network namespace	Customer-defined via Kubernetes manifests (YAML)
Services	Stable network endpoint exposing pods internally or externally (ClusterIP, NodePort, LoadBalancer)	Customer-defined; LoadBalancer type creates Azure Load Balancer
Ingress	HTTP/HTTPS routing rules for external traffic; integrates with Azure Application Gateway	Customer-deployed; AGIC add-on available
Azure CNI	Advanced networking giving each pod a routable IP from the VNet subnet	Configured at cluster creation

2.5 Azure Functions (Serverless Compute)

Azure Functions is a serverless compute service that lets you run small, event-triggered code units (functions) without provisioning or managing any servers. You pay only for the compute time consumed—billed per execution and GB-second of memory.

Functions are ideal for event-driven workloads such as processing messages from Service Bus, reacting to blob uploads, handling HTTP webhooks, and building lightweight APIs. They integrate with 200+ Azure and third-party services via triggers and bindings.

Hosting Plan	Scale	Cold Start	Max Timeout	VNet Integration	Best For
Consumption	Automatic, per-event	Yes (~1-3s)	10 minutes	No (Premium required)	Infrequent, event-driven workloads
Premium	Automatic with pre-warmed instances	No (pre-warmed)	Unlimited	Yes (full)	Performance-critical, VNet-required functions
Dedicated (App Service)	Manual or auto-scale like App Service	No	Unlimited	Yes	Predictable load, co-located with web apps
Container Apps	KEDA-based scale to zero	Configurable	Unlimited	Yes	Containerized functions, microservices architectures

Advanced Concept: Durable Functions

Durable Functions: An extension of Azure Functions that enables stateful workflows in a serverless environment. Supports fan-out/fan-in, chaining, human interaction, and long-running processes using orchestrator and activity functions.

Chapter 3: Azure Storage Services

Azure Storage is Microsoft's cloud storage solution providing highly available, massively scalable, durable, and secure storage for a wide variety of data objects. Azure Storage is the foundation for most Azure services and supports data at exabyte scale, with 11 nines (99.999999999%) durability for locally redundant storage.

3.1 Azure Storage Account

An Azure Storage Account is the top-level resource and unique namespace for Azure Storage. Every object stored in Azure Storage has an address that includes the unique account name, forming URLs like myaccount.blob.core.windows.net/container/blob.

Account Type	Supported Services	Performance Tier	Supported Redundancy	Use Case
Standard General Purpose v2 (GPv2)	Blob, Files, Queue, Table, Data Lake Gen2	Standard (HDD-backed)	LRS, GRS, RA-GRS, ZRS, GZRS, RA-GZRS	Default choice for most scenarios
Premium Block Blobs	Block Blobs and Append Blobs only	Premium (SSD-backed)	LRS, ZRS	Low-latency blob scenarios, AI/ML pipelines
Premium File Shares	Azure Files only	Premium (SSD-backed)	LRS, ZRS	High-performance SMB/NFS file shares
Premium Page Blobs	Page Blobs only	Premium (SSD-backed)	LRS only	VM unmanaged disks (legacy)

3.2 Azure Blob Storage

Azure Blob Storage is optimized for storing massive amounts of unstructured data—text or binary—such as images, videos, audio files, documents, log files, and backups. Blob storage is accessible via HTTPS/REST, Azure SDKs, Azure CLI, and Storage Explorer.

Blob Type	Max Size	Use Case	Notes
Block Blob	Up to 190.7 TB	General-purpose: images, documents, videos, backups, log files	Composed of up to 50,000 blocks; supports parallel upload
Append Blob	Up to 195 GB	Log files, audit trails, telemetry streams	Optimized for append-only operations; cannot modify existing blocks
Page Blob	Up to 8 TB	Azure VM disks (VHDs), random read/write workloads	Supports random read/write access; 512-byte pages

Blob Storage Access Tiers

Azure Blob Storage offers tiered storage to optimize costs based on how frequently data is accessed. Data can be automatically moved between tiers using Lifecycle Management policies.

Access Tier	Access Frequency	Storage Cost	Access/Operation Cost	Retrieval Time	Min Storage Duration
Hot	Frequent (multiple times/day)	Highest per GB	Lowest per operation	Immediate	None
Cool	Infrequent (once a month)	Lower per GB	Higher per operation	Immediate	30 days
Cold	Rarely accessed (once every 90 days)	Even lower per GB	Even higher per operation	Immediate	90 days
Archive	Very rarely accessed (once a year+)	Lowest per GB	Highest per operation	1–15 hours (rehydration)	180 days

Lifecycle Management Best Practice

Lifecycle Management: Define rules to automatically transition blobs between tiers based on age or last modification time.
Example Rule: Move to Cool after 30 days → Move to Archive after 90 days → Delete after 365 days.
This can reduce storage costs by 60–80% for data with predictable access patterns.

3.3 Azure Files

Azure Files offers fully managed file shares in the cloud, accessible via the industry-standard Server Message Block (SMB 2.1/3.0) and Network File System (NFS 4.1) protocols. Azure file shares can be mounted concurrently by Windows, Linux, and macOS—both cloud and on-premises deployments.

- Lift-and-shift: Migrate on-premises file servers to Azure without changing application code.
- Cloud-native: Replace UNC paths in cloud applications with Azure file shares mounted as network drives.
- Azure File Sync: Extend on-premises Windows Server with Azure Files to enable tiering, disaster recovery, and multi-site sync.
- Snapshots: Point-in-time snapshots of entire file shares, stored efficiently (only changes).

3.4 Storage Redundancy Options

Azure Storage replicates data to protect against hardware failures, network outages, and datacenter disasters. You choose a redundancy level when creating a storage account, trading cost for durability and availability.

Redundancy Option	Copies	Scope	Read Access in Failure	SLA	Best For
LRS (Locally Redundant)	3 synchronous copies	Single datacenter, same region	No failover access	99.999999999% durability	Low-cost dev/test, data easily recreated
ZRS (Zone Redundant)	3 synchronous copies	3 availability zones, same region	Access continues if one zone fails	99.9999999999% durability	High availability in-region, compliance

Redundancy Option	Copies	Scope	Read Access in Failure	SLA	Best For
GRS (Geo-Redundant)	3 LRS + 3 async copies	Primary + secondary region	Only after failover (RA-GRS: yes)	99.99999999999999% durability	Regional disaster recovery
GZRS (Geo-Zone Redundant)	3 ZRS + 3 async copies	3 zones in primary + secondary region	Only after failover (RA-GZRS: yes)	99.99999999999999% durability	Maximum durability and availability

Chapter 4: Azure Networking

Azure Networking forms the backbone connecting all Azure services and linking Azure cloud environments to on-premises datacenters and the public internet. A thorough understanding of Azure networking is essential for building secure, scalable, and highly available architectures.

4.1 Azure Virtual Network (VNet)

Azure Virtual Network (VNet) is the fundamental building block for private networking in Azure. A VNet enables Azure resources such as Virtual Machines, App Services, and databases to securely communicate with each other, the internet, and on-premises networks. VNets are isolated from each other by default, providing network-level isolation as a security boundary.

VNet Concept	Description	Key Detail
Address Space	CIDR block defining the IP range for the VNet (e.g., 10.0.0.0/16)	Can add multiple address spaces; must not overlap with peered VNets
Subnet	Subdivision of VNet address space for organizing and isolating resources	Resources in a subnet get IPs from that subnet's range; NSGs apply at subnet level
VNet Peering	Low-latency, high-bandwidth connection between two VNets (same or different regions)	Traffic stays on Azure backbone; no encryption overhead; not transitive by default
Route Tables	Custom user-defined routes (UDR) to override Azure's default routing behavior	Force traffic through Network Virtual Appliance (NVA) or Azure Firewall
Service Endpoints	Extend VNet identity to Azure PaaS services over the Azure backbone network	Removes exposure to public internet; supported for Storage, SQL, Key Vault, etc.
Private Endpoints	Private IP address in your VNet for an Azure PaaS service (like a NIC in your VNet)	Fully private connectivity; no public IP needed; DNS resolution to private IP

4.2 Network Security Groups (NSG)

A Network Security Group (NSG) contains a list of security rules that allow or deny inbound or outbound network traffic to Azure resources. NSGs can be applied to individual subnets or individual virtual machine NICs. Rules are evaluated in priority order; the first matching rule wins.

NSG Property	Allowed Values	Description
Priority	100 – 4096	Lower number = higher priority. Rules processed in ascending priority order. First match wins.
Source / Destination	IP, CIDR, Service Tag, Application Security Group	Defines where traffic originates (source) or where it's going (destination)

NSG Property	Allowed Values	Description
Port / Port Range	Single port (443), range (1024-65535), or * (all)	The TCP/UDP port(s) the rule applies to
Protocol	TCP, UDP, ICMP, ESP, AH, or Any (*)	Network protocol the rule applies to
Action	Allow or Deny	Whether to permit or block matching traffic
Service Tags	AzureLoadBalancer, Internet, VirtualNetwork, AzureCloud, Storage, Sql, etc.	Simplified groupings of Azure service IP ranges, automatically maintained by Microsoft

⚠ Important: Default NSG Rules

Default NSG Rules: Every NSG has 3 inbound and 3 outbound default rules (priority 65000–65500) that cannot be deleted:

- Allow inbound VirtualNetwork traffic (VNet-to-VNet)
- Allow inbound Azure Load Balancer health probes
- Deny all other inbound traffic
- Allow outbound VirtualNetwork traffic
- Allow outbound internet traffic
- Deny all other outbound traffic

4.3 Azure Load Balancer

Azure Load Balancer operates at Layer 4 (Transport layer) of the OSI model and distributes inbound TCP/UDP flows among healthy backend instances. It enables high availability and network performance for applications.

Load Balancer Type	Tier	Frontend IP	Use Case	Availability Zones
Public Load Balancer (Standard)	Standard	Public IP address	Internet-facing apps; distribute traffic from internet to VMs	Zone-redundant or zonal
Internal Load Balancer (Standard)	Standard	Private IP (from VNet)	Internal apps; distribute traffic between tiers (web→app)	Zone-redundant or zonal
Basic Load Balancer	Basic (legacy)	Public or Private	Simple dev/test scenarios only; no SLA, no zones	Not supported

4.4 Azure Application Gateway & WAF

Azure Application Gateway is a Layer 7 (Application layer) web traffic load balancer that provides advanced traffic management capabilities including SSL/TLS termination, URL-based routing, session affinity, and Web Application Firewall (WAF).

- SSL/TLS Termination: Offloads SSL decryption at the gateway, reducing CPU burden on backend servers.
- URL-Based Routing: Route traffic to different backend pools based on URL path (e.g., /images → image servers, /api → API servers).

- Multi-Site Hosting: Host multiple websites behind a single Application Gateway using host header routing.
- WAF (Web Application Firewall): Protects against OWASP Top 10 vulnerabilities including SQL injection, XSS, and CSRF.
- Autoscaling: Automatically adjusts capacity based on traffic load patterns (Standard v2 and WAF v2 SKUs).
- Health Probes: Custom HTTP/HTTPS probes to monitor backend pool health and automatically remove unhealthy instances.

4.5 Azure DNS

Azure DNS is a hosting service for DNS domains providing name resolution using Microsoft Azure infrastructure. By hosting domains in Azure, you manage DNS records using the same credentials, APIs, and billing as other Azure services. Azure DNS supports both public DNS zones (for internet-facing domains) and private DNS zones (for VNet-only name resolution).

4.6 Hybrid Connectivity

Connecting on-premises networks to Azure is a fundamental requirement for most enterprise cloud adoption scenarios. Azure provides two primary options for hybrid connectivity: VPN Gateway and Azure ExpressRoute.

Connectivity Option	Type	Bandwidth	Latency	SLA	Cost	Best For
VPN Gateway (Site-to-Site)	Encrypted IPsec/IKE tunnel over public internet	Up to 10 Gbps (VpnGw5AZ)	Variable (internet-dependent)	99.9% – 99.99%	Lower cost	Branch office connectivity, backup path for ExpressRoute
VPN Gateway (Point-to-Site)	Individual device VPN over internet (SSTP/OpenVPN/IKEv2)	Up to 100 Mbps	Variable	99.9% – 99.99%	Low cost	Remote workers accessing Azure resources
ExpressRoute	Private dedicated circuit via connectivity provider	50 Mbps – 100 Gbps	Predictable, low (<10ms common)	99.95%	Higher cost	Mission-critical, high-bandwidth, consistent latency
ExpressRoute Global Reach	Connect multiple on-premises sites via Azure backbone	Same as ExpressRoute circuits	Low, consistent	99.95%	Additional cost	Connecting branch offices globally through Azure

Chapter 5: Azure Identity & Security

Security is a top priority in Azure. Microsoft employs a shared responsibility model where both Microsoft and the customer are responsible for different layers of security depending on the service model (IaaS/PaaS/SaaS). Azure provides a comprehensive, defense-in-depth security portfolio spanning identity, network, data, and application layers.

5.1 Microsoft Entra ID (formerly Azure Active Directory)

Microsoft Entra ID is Microsoft's cloud-based identity and access management (IAM) service. It is the backbone of authentication and authorization for Microsoft 365, Azure, thousands of SaaS applications, and custom enterprise applications. Entra ID replaced the legacy "Azure Active Directory" branding in 2023.

Unlike Windows Server Active Directory (which is LDAP/Kerberos based), Entra ID is built for the internet, using modern protocols: OAuth 2.0, OpenID Connect (OIDC), and SAML 2.0.

Entra ID Concept	Definition	Key Detail
Tenant	A dedicated, isolated instance of Entra ID representing an organization	Each tenant has a unique domain (e.g., contoso.onmicrosoft.com) and tenant ID (GUID)
User	A person's identity managed in Entra ID; can be Member or Guest	Supports cloud-only users, synchronized from on-prem AD, or B2B guest users
Group	Collection of users (or devices/apps) for simplified access management	Security Groups (access control) and Microsoft 365 Groups (collaboration)
Application	Any software registered in Entra ID to use it for authentication	Represented by App Registration (developer config) + Enterprise Application (tenant instance)
Service Principal	Non-human security identity for an application, service, or automated process	Created automatically when an App Registration is used in a tenant
Managed Identity	Auto-managed service principal for Azure resources; no credentials to store or rotate	System-assigned (1:1 with resource) or User-assigned (shareable across resources)

Entra ID License Tiers

License	Key Features
Free	User management, SSO (up to 10 apps), basic security reports, B2B collaboration
Microsoft Entra ID P1	Conditional Access, hybrid identity, self-service password reset, dynamic groups, PIM (basic)
Microsoft Entra ID P2	Everything in P1 + Identity Protection (risk-based Conditional Access), Privileged Identity Management (full PIM)

5.2 Role-Based Access Control (RBAC)

Azure RBAC is an authorization system built on Azure Resource Manager that provides fine-grained access management to Azure resources. RBAC follows the principle of least privilege—granting users only the permissions they need to perform their job functions.

A Role Assignment binds a Security Principal (who) to a Role Definition (what permissions) at a specific Scope (where it applies).

Built-in Role	Permissions	Typical Assignee
Owner	Full access to all resources including ability to delegate access (assign roles)	Subscription owners, resource group owners
Contributor	Create and manage all resources but cannot grant access to others	Application teams, DevOps engineers
Reader	View all resources but cannot make any changes	Auditors, stakeholders needing visibility
User Access Administrator	Manage user access to Azure resources (grant/revoke role assignments)	Identity/security administrators
Custom Roles	Define exact permissions needed using action strings (e.g., Microsoft.Compute/virtualMachines/start/action)	When built-in roles are too broad or too narrow

RBAC Scope Inheritance

RBAC Scope Hierarchy: Management Group → Subscription → Resource Group → Individual Resource
Roles assigned at a higher scope are inherited by all child scopes. A Contributor at the Subscription level has Contributor access to ALL resource groups and resources within that subscription.

5.3 Azure Key Vault

Azure Key Vault is a cloud-based HSM (Hardware Security Module) and secrets management service. It centralizes application secrets, cryptographic keys, and certificates—eliminating the dangerous practice of embedding sensitive information in application code or configuration files.

Key Vault Object Type	Description	Key Capabilities
Secrets	Arbitrary strings: passwords, connection strings, API keys, tokens, certificates	Versioned; soft-delete and purge protection; access logged in Azure Monitor
Keys	Cryptographic keys for encryption/decryption operations (RSA 2048/4096, EC P-256/P-384)	HSM-backed keys never leave HSM in plaintext; supports Bring Your Own Key (BYOK)
Certificates	X.509 SSL/TLS certificates with full lifecycle management	Automatic renewal with DigiCert or Entrust CAs; integrated with App Service, AGIC, API Management

5.4 Microsoft Defender for Cloud

Microsoft Defender for Cloud is a Cloud Security Posture Management (CSPM) and Cloud Workload Protection Platform (CWPP). It continuously assesses your security posture, provides actionable recommendations, and offers threat protection for hybrid and multi-cloud workloads.

- Secure Score: A numeric score (0–100%) representing your current security posture; actionable recommendations to improve it.
- Security Policies: Azure Policy-based controls mapped to compliance standards (CIS, NIST, PCI-DSS, ISO 27001).
- Defender Plans: Workload-specific threat protection for VMs, containers, storage, databases, Key Vault, DNS, and more.
- Regulatory Compliance Dashboard: Track compliance against industry standards and regulations.
- Just-In-Time VM Access: Reduce attack surface by opening VM management ports (RDP/SSH) only when needed, for specific IPs, for limited time.

5.5 Zero Trust Security Model

The Zero Trust model is a security framework that assumes breach—treating every user, device, and network connection as untrusted regardless of whether it is inside or outside the corporate network perimeter. Azure's security services are designed to support Zero Trust implementation across identity, endpoints, applications, data, infrastructure, and networks.

Zero Trust Principle	Description	Azure Implementation
Verify Explicitly	Always authenticate and authorize based on all available data points (identity, location, device, service, workload, data classification, anomalies)	Entra ID Conditional Access, MFA, Identity Protection
Use Least Privilege Access	Limit user access with Just-In-Time (JIT) and Just-Enough-Access (JEA) policies; protect data and productivity	Azure RBAC, PIM, JIT VM Access, Entra ID P2
Assume Breach	Minimize blast radius; segment access; verify end-to-end encryption; use analytics to gain visibility and drive threat detection	Microsoft Sentinel, Defender for Cloud, Azure Monitor, Private Endpoints

Chapter 6: Azure Databases & Data Services

Azure provides one of the most comprehensive portfolios of fully managed database services covering relational, NoSQL, in-memory, graph, and analytical data stores. These services handle patching, backups, high availability, and scaling automatically, allowing teams to focus on application development.

6.1 Azure SQL Database

Azure SQL Database is a fully managed relational database service based on the latest stable version of Microsoft SQL Server Database Engine. It provides built-in high availability (99.99% SLA), automated backups with point-in-time restore (up to 35 days), intelligent performance optimization via Query Performance Insights, and seamless scalability.

Feature	Azure SQL Database	SQL Managed Instance	SQL Server on Azure VM
Deployment Model	PaaS (fully managed)	PaaS (nearly fully managed)	IaaS (customer manages OS+SQL)
SQL Server Compatibility	~99% (some limitations)	~100% (near-full compat.)	100% (exact version control)
Built-in HA	Yes – 99.99% SLA	Yes – 99.99% SLA	Manual setup required
Automated Backups	Yes – up to 35 days PITR	Yes – up to 35 days PITR	Manual (SQL Agent or Azure Backup)
Cross-DB Queries	Elastic queries only	Yes – full support	Yes – full support
Linked Servers	Not supported	Yes – supported	Yes – supported
SQL Agent Jobs	Elastic Jobs (external service)	Yes – native SQL Agent	Yes – native SQL Agent
VNet Integration	Private Endpoint or Service Endpoint	Native VNet injection (fully private)	Full VNet via VM NIC
Best For	New cloud apps, SaaS apps, microservices	Lift-and-shift of on-prem SQL Server	Full control, legacy apps, specific SQL version

6.2 Azure Cosmos DB

Azure Cosmos DB is a fully managed, globally distributed, multi-model NoSQL database service. It guarantees single-digit millisecond (< 10ms) response times at the 99th percentile at any scale, and offers five consistency levels, multiple API options, and automatic multi-region writes.

Cosmos DB API	Data Model	Query Language	Migrate From	Best Use Case
NoSQL (Core SQL)	Document (JSON)	SQL-like (SELECT, WHERE, JOIN)	Custom apps	New cloud-native apps needing flexible schemas
MongoDB	Document (BSON)	MongoDB Query Language (MQL)	MongoDB	Migrate existing MongoDB apps without code changes

Cosmos DB API	Data Model	Query Language	Migrate From	Best Use Case
Cassandra	Wide-column / Column-family	Cassandra Query Language (CQL)	Apache Cassandra	Time-series data, IoT telemetry, high-write workloads
Gremlin	Graph (vertices + edges)	Gremlin graph traversal language	Apache TinkerPop	Social networks, recommendation engines, fraud detection
Table	Key-value pairs	OData / Table REST API	Azure Table Storage	Simple key-value lookups, schema-less data
PostgreSQL	Relational (distributed, Citus)	SQL (PostgreSQL dialect)	PostgreSQL	Horizontally sharded relational workloads

Cosmos DB Consistency Levels

Cosmos DB offers five consistency levels on a spectrum between perfect consistency (strong) and best performance (eventual). You choose based on your application's balance of availability, latency, throughput, and data freshness requirements.

Consistency Level	Guarantee	Availability	Read Latency	Throughput	Use Case
Strong	Reads always return the most recent committed write (linearizability)	Lower (less available in partitioned network)	Higher	Lower	Financial transactions requiring absolute accuracy
Bounded Staleness	Reads lag behind writes by at most K operations or T time interval	High	Low-Medium	Medium	Apps needing mostly current data with bounded lag
Session	Within a session: reads your own writes; consistent prefix; monotonic reads	High	Low	High	Default; shopping carts, user profiles, social feeds
Consistent Prefix	Reads never see out-of-order writes; all writes visible in order	High	Low	High	Distributed counters, social media timelines
Eventual	No ordering guarantee; eventually all replicas converge	Highest	Lowest	Highest	Non-ordered apps: DNS, leaderboards, likes/favorites

6.3 Azure Synapse Analytics

Azure Synapse Analytics is an unlimited analytics service that unifies enterprise data warehousing and Big Data analytics. It provides a single interface to query data using either serverless SQL (pay-per-query) or dedicated SQL pools (provisioned capacity), run Apache Spark for big data processing and machine learning, and orchestrate data pipelines.

- Synapse SQL: Serverless (ad-hoc queries on Data Lake) or Dedicated SQL Pool (provisioned DWU-based data warehouse).
- Apache Spark Pool: Distributed data processing and machine learning using PySpark, Scala, SQL, or .NET.
- Synapse Pipelines: Code-free data integration (ADF-based) for ETL/ELT workflows.
- Synapse Link: Near-real-time analytics on operational data from Cosmos DB, SQL Server, Dataverse without ETL.
- Azure Data Lake Storage Gen2: Integrated, hierarchical namespace storage for petabyte-scale analytics data.

Chapter 7: Azure AI & Machine Learning

Azure provides one of the most comprehensive enterprise AI platforms, enabling developers and data scientists to build, deploy, and scale AI models and applications. From pre-built cognitive APIs to full MLOps pipelines and foundation model access via Azure OpenAI, Azure covers the complete AI development lifecycle.

7.1 Azure AI Services (Cognitive Services)

Azure AI Services (formerly Azure Cognitive Services) provides pre-built AI capabilities through simple REST APIs and SDKs, enabling developers to add intelligent features without requiring deep AI or ML expertise.

Category	Service	Key Capability
Vision	Azure AI Vision (Computer Vision)	OCR, image description, object detection, brand recognition, spatial analysis
Vision	Azure AI Face	Face detection, verification, identification, liveness detection (anti-spoofing)
Vision	Azure AI Custom Vision	Train custom image classification and object detection models with few images
Language	Azure AI Language	Sentiment analysis, named entity recognition, summarization, key phrase extraction
Language	Azure AI Translator	Neural machine translation across 100+ languages; custom translation models
Language	Azure AI Document Intelligence	Extract structured data from forms, invoices, receipts, ID documents, contracts
Speech	Azure AI Speech (STT)	Real-time speech-to-text; custom acoustic/language models; speaker diarization
Speech	Azure AI Speech (TTS)	Neural text-to-speech with natural prosody; custom neural voice creation
Decision	Azure AI Anomaly Detector	Detect anomalies in time-series data (metrics, telemetry) using ML
Decision	Azure AI Content Safety	Detect and filter harmful content: hate speech, violence, sexual content, self-harm
Search	Azure AI Search	Cloud search-as-a-service with AI enrichment pipeline, vector search, semantic ranking

7.2 Azure Machine Learning

Azure Machine Learning (Azure ML) is an enterprise-grade platform for the end-to-end machine learning lifecycle. It provides tools for data preparation, model training (manual and AutoML), experiment tracking, model registry, deployment, and monitoring—enabling MLOps best practices at scale.

Azure ML Component	Description	Key Feature
Workspace	Top-level resource containing all Azure ML assets; connects compute, data, and experiments	Hub for collaboration; RBAC-controlled; region-based
Datasets / Data Assets	Registered data references (files, folders, tables) in Azure Storage or Data Lake	Versioned; lineage tracked; supports tabular and file data
Compute Clusters	Managed, on-demand VM clusters for training jobs; scale to zero when idle	Supports CPU and GPU VMs; low-priority option for cost savings
AutoML	Automated model selection and hyperparameter tuning; tries dozens of algorithms automatically	Supports classification, regression, forecasting, NLP, CV
Designer	Visual drag-and-drop interface for building ML pipelines without writing code	Pre-built modules; export to Python for customization
MLflow Integration	Open-source experiment tracking, model registry, and deployment standard	Log metrics, parameters, artifacts; compare runs
Model Registry	Central repository for versioned ML models with metadata and stage management	Staging → Production lifecycle; champion/challenger tracking
Managed Endpoints	Deploy models to real-time (online) or batch inference endpoints	Kubernetes-backed; blue/green deployment; traffic splitting

7.3 Azure OpenAI Service

Azure OpenAI Service provides enterprise-grade REST API access to OpenAI's most powerful language models including GPT-4o, GPT-4, DALL-E 3, and Whisper. Unlike the public OpenAI API, Azure OpenAI includes enterprise capabilities: private networking (VNet integration), responsible AI content filtering, data residency, no training on customer data, and Azure RBAC.

Model / Feature	Capability	Common Use Case
GPT-4o (multimodal)	Processes text, images, audio; most capable and efficient GPT-4 class model	Advanced reasoning, code generation, document analysis with images
GPT-4 Turbo	Large context window (128K tokens), function calling, JSON mode	Long-document analysis, complex reasoning, structured data extraction
GPT-3.5 Turbo	Fast, cost-efficient language model; 16K context window	Chatbots, Q&A, content classification, summarization at scale
DALL-E 3	Generate high-quality images from natural language text descriptions	Creative assets, product visualization, marketing content

Model / Feature	Capability	Common Use Case
Whisper	Speech-to-text transcription supporting 100+ languages; high accuracy	Meeting transcription, caption generation, voice interfaces
Text Embeddings	Convert text to high-dimensional vector representations	Semantic search, RAG (Retrieval-Augmented Generation), similarity
Assistants API	Stateful AI assistants with file retrieval, code interpreter, function calling	Build production AI assistants without managing conversation state

Chapter 8: Azure DevOps & Governance

Azure DevOps provides an integrated suite of developer tools for planning, collaborating, and delivering software continuously. Azure governance services complement DevOps by providing policy enforcement, cost control, compliance management, and organizational hierarchy across Azure subscriptions.

8.1 Azure DevOps Services

Azure DevOps is a SaaS platform offering five integrated services for software delivery teams, from idea to production deployment.

Azure DevOps Service	Description	Key Capability
Azure Boards	Agile project management with Kanban boards, backlogs, sprints, and work item tracking	Customizable work item types; GitHub integration; velocity/burndown charts
Azure Repos	Git repositories with pull request workflow, branch policies, and code review	Branch protection rules; required reviewers; build validation on PR
Azure Pipelines	CI/CD automation for building, testing, and deploying to any platform or cloud	YAML pipelines as code; multi-stage; environments; approvals; gates
Azure Artifacts	Universal package management for NuGet, npm, Maven, Python, and Universal packages	Upstream sources; package retention policies; feed-scoped permissions
Azure Test Plans	Manual and exploratory testing tools with test case management	Test runs, test results, defect tracking; browser extension for exploratory testing

8.2 CI/CD Pipeline Architecture

A well-designed CI/CD pipeline automates the path from code commit to production deployment, enforcing quality gates, security scanning, and approval workflows at each stage.

Pipeline Stage	Activities	Key Tools
Source (Trigger)	Code push to feature branch, pull request creation triggers pipeline run	Azure Repos, GitHub; branch policies; PR validation builds
Build (CI)	Compile code, restore packages, run unit tests, publish test results, build container image	dotnet build/test, Maven, npm, Docker; Azure Pipelines YAML
Static Analysis	SAST (static application security testing), code coverage, dependency scanning	SonarCloud, Checkmarx, WhiteSource Bolt, GitHub Advanced Security
Artifact Publishing	Push build outputs (binaries, container images, packages) to artifact repositories	Azure Artifacts, Azure Container Registry (ACR), Artifactory

Pipeline Stage	Activities	Key Tools
Deploy to Dev	Automated deployment to development environment; run integration and smoke tests	Azure Pipelines environments; Helm charts; ARM/Bicep templates
Deploy to Staging	Deploy to staging environment; run full regression suite; performance tests	Deployment slots (App Service); approval gates; traffic validation
Deploy to Production	Controlled production rollout with manual approval; blue/green or canary strategy	Traffic Manager, Azure Front Door; rollback capability; monitoring alerts

8.3 Azure Governance

Azure governance tools provide the organizational structure, policy enforcement, and cost management capabilities needed to operate Azure at enterprise scale with compliance and accountability.

Governance Tool	Purpose	Key Capability
Management Groups	Organize subscriptions into a hierarchy for unified policy and access management across the enterprise	Up to 6 levels deep; policies/RBAC assigned here propagate to all child subscriptions
Azure Policy	Create, assign, and manage policies to enforce compliance rules and effects on resources	Effects: Deny, Audit, Append, Modify, DeployIfNotExists; Initiative = collection of policies
Azure Blueprints	Define repeatable sets of Azure resources, policies, RBAC assignments, and ARM templates	Version-controlled; assign to subscriptions; track compliance; updates propagate
Resource Tags	Key-value metadata pairs for organizing, searching, and cost allocation of Azure resources	Required tags enforced via Azure Policy; used for cost center chargeback in Cost Management
Azure Cost Management	Monitor, allocate, analyze, and optimize Azure costs with budgets and cost alerts	Cost analysis, budgets, alerts, recommendations, export for chargeback; FinOps practices
Azure Advisor	Personalized, actionable recommendations across 5 categories based on usage and configuration	Reliability, Security, Performance, Cost, Operational Excellence recommendations
Azure Resource Graph	Query and explore all your Azure resources across subscriptions using Kusto Query Language (KQL)	Snapshot queries at scale; governance dashboards; compliance reporting

8.4 Azure Monitor & Observability

Azure Monitor is the unified observability platform for Azure, collecting metrics, logs, and traces from Azure resources, applications, and on-premises environments. It enables proactive monitoring, intelligent alerting, and operational insights.

Azure Monitor Component	Description	Typical Use
Metrics	Numerical time-series data emitted by Azure resources (CPU%, requests/sec, latency)	Dashboards, autoscale triggers, alert rules
Log Analytics (Logs)	Log Kusto (KQL) queries against structured log data from resources, agents, apps	Root cause analysis, complex queries, compliance reports
Application Insights	APM (Application Performance Monitoring) SDK for web apps; traces, dependencies, exceptions	End-to-end tracing, performance profiling, availability tests
Azure Monitor Alerts	Trigger notifications or actions based on metrics/log conditions	Email/SMS/webhook/PagerDuty; action groups; auto-remediation runbooks
Workbooks	Interactive report and dashboard templates using metrics + logs + ARM data	Custom operational dashboards; shared across teams
Azure Service Health	Personalized alerts for Azure service issues, planned maintenance, health advisories	Know when Azure problems affect your specific resources
Microsoft Sentinel	Cloud-native SIEM and SOAR; collect security logs, detect threats, respond automatically	Security operations, threat hunting, compliance reporting

Chapter 9: Azure Architecture & Well-Architected Framework

The Azure Well-Architected Framework (WAF) provides a set of guiding principles, best practices, and assessment tools to help architects design workloads that are reliable, secure, cost-effective, operationally excellent, and performant. It is organized around five pillars.

9.1 The Five Pillars of the Well-Architected Framework

Pillar	Core Question	Key Principles	Azure Tools
Reliability	Will this workload meet its availability and recovery targets?	Design for failure; use availability zones/regions; test failure scenarios; define RTO/RPO; circuit breakers	Azure Site Recovery, Azure Backup, Traffic Manager, Azure Monitor, Chaos Studio
Security	How do we protect the workload against threats?	Defense in depth; least privilege; zero trust; encrypt data at rest and in transit; threat detection	Entra ID, Azure RBAC, Key Vault, Defender for Cloud, Azure DDoS Protection, WAF
Cost Optimization	How do we deliver maximum value at minimum cost?	Right-size resources; use reservations; auto-shutdown dev/test; delete unused resources; PaaS over IaaS	Azure Cost Management, Azure Advisor, Reserved Instances, Azure Hybrid Benefit
Operational Excellence	How do we operate and continuously improve the workload?	Infrastructure as Code; CI/CD pipelines; immutable deployments; monitoring; runbooks; blameless postmortems	Azure DevOps, Azure Monitor, Azure Policy, ARM/Bicep/Terraform, Azure Automation
Performance Efficiency	Does the workload scale and perform as demand changes?	Scale horizontally; cache frequently accessed data; CDN for static content; async processing; performance testing	Azure Autoscale, Azure Cache for Redis, Azure CDN, Application Gateway, Load Balancer

9.2 Common Azure Architecture Patterns

N-Tier Architecture

The N-Tier (or multi-tier) architecture separates application concerns into distinct layers (presentation, application/business logic, data), each hosted on separate infrastructure. This enables independent scaling, security segmentation, and clear separation of concerns.

Tier	Responsibility	Azure Service	Security Boundary
Presentation (Tier 1)	User interface; HTML/CSS/JS rendering; static content delivery	Azure Static Web Apps, App Service, Azure CDN	WAF + Application Gateway; NSG on subnet

Tier	Responsibility	Azure Service	Security Boundary
Application (Tier 2)	Business logic, API processing, orchestration, authentication	App Service, AKS, Azure Functions, Azure API Management	NSG restricting traffic from Tier 1 subnet only
Data (Tier 3)	Persistent data storage, caching, messaging	Azure SQL, Cosmos DB, Azure Cache for Redis, Service Bus	Private Endpoints; no public internet access; Tier 2 access only

Microservices Architecture

Microservices decompose applications into small, independently deployable services that communicate via APIs or messaging. Each service owns its data store and can be developed, deployed, and scaled independently. AKS is the preferred Azure platform for microservices.

- API Gateway Pattern: Azure API Management or Application Gateway acts as the single entry point, handling routing, auth, rate limiting, and transformation.
- Service-to-Service Communication: Synchronous via REST/gRPC (internal services using Kubernetes ClusterIP) or asynchronous via messaging (Azure Service Bus, Event Hubs, Event Grid).
- Data Isolation: Each microservice has its own dedicated database. No shared database between services.
- Sidecar Pattern: Dapr (Distributed Application Runtime) provides service discovery, state management, pub/sub, and observability as sidecars in AKS.
- Health Probes: Kubernetes liveness and readiness probes ensure traffic only routes to healthy pods.

9.3 Disaster Recovery Concepts

DR Concept	Definition	Target	Azure Service / Approach
RTO (Recovery Time Objective)	Maximum acceptable time for an application to be offline after a disaster	As low as seconds (Active-Active) to hours (backup restore)	Azure Site Recovery, Traffic Manager, Azure Front Door
RPO (Recovery Point Objective)	Maximum acceptable data loss measured in time (how far back can we restore?)	Near-zero (synchronous replication) to hours (daily backup)	Azure Backup, SQL Geo-replication, Cosmos DB multi-region write
Active-Active	Both primary and secondary regions simultaneously serve live traffic; instant failover	RTO: ~0 seconds; RPO: ~0 seconds	Azure Traffic Manager (performance routing), Azure Front Door
Active-Passive (Hot Standby)	Secondary region is running and ready but receives no production traffic; fast failover	RTO: minutes; RPO: seconds (async replication)	Azure Site Recovery, SQL Failover Groups
Active-Passive (Warm Standby)	Secondary infrastructure is partially provisioned; needs startup time	RTO: tens of minutes; RPO: minutes	ARM templates for rapid deployment; geo-redundant backups

Chapter 10: Azure Pricing, SLAs & Certification Guide

10.1 Azure Pricing Models

Azure offers flexible pricing models to match different usage patterns, commitment levels, and organizational policies. Understanding these models is essential for cost optimization and accurate cloud budgeting.

Pricing Model	Description	Typical Savings	Best For
Pay-As-You-Go (PAYG)	Pay per second/hour for actual usage with no upfront commitment or cancellation fee	Baseline (no discount)	Variable workloads, dev/test, new projects
Reserved Instances (RI)	1 or 3-year commitment for specific VM size and region; pay upfront or monthly	40–72% vs PAYG	Stable production workloads with predictable demand
Spot / Low Priority VMs	Bid on unused Azure capacity; evictable with 30-second notice	Up to 90% vs PAYG	Fault-tolerant batch jobs, rendering, testing
Azure Hybrid Benefit	Apply existing on-premises Windows Server and SQL Server licenses to Azure VMs	40–55% savings on license costs	Organizations with existing Microsoft volume licensing
Dev/Test Pricing	Discounted rates for non-production workloads on Visual Studio subscriptions	Various (e.g., no Windows license cost on VMs)	Development and test environments only
Savings Plans (Compute)	1 or 3-year commitment to hourly spend (not specific VM SKU); flexible across VM families	15–65% vs PAYG	Dynamic workloads needing flexibility across VM sizes/regions
Azure Free Account	12 months of popular services free + $200 credit first 30 days + 55+ always-free services	100% free during trial period	Learning, POC, individual developers

10.2 Total Cost of Ownership (TCO) & Cost Optimization

The Azure TCO Calculator estimates cost savings when migrating on-premises workloads to Azure. It accounts for server hardware, software licenses, electricity, datacenter space, network, and IT labor costs. Most organizations report 30–50% total cost reduction over 5 years.

☑ Cost Optimization Best Practices

1. Right-size Resources: Use Azure Advisor to identify underutilized VMs and downsize or change tier.
2. Reserved Instances: For steady-state workloads, 1 or 3-year RIs save up to 72% over PAYG.
3. Auto-Shutdown: Schedule dev/test VMs to stop outside business hours (saves ~70% of compute cost).
4. Delete Unused Resources: Remove orphaned disks, public IPs, empty resource groups, and idle services.
5. Azure Hybrid Benefit: Apply Windows Server and SQL Server licenses to save 40–55%.

6. Storage Tiering: Move infrequently accessed blobs to Cool/Archive tier using Lifecycle Management.
7. PaaS vs IaaS: PaaS services often cost less at scale due to shared infrastructure and no OS licensing.
8. Budgets and Alerts: Set Azure Cost Management budgets with email alerts at 80% and 100% thresholds.

10.3 Service Level Agreements (SLAs)

An SLA defines Microsoft's commitment to uptime and connectivity for each Azure service. Failing to meet the SLA results in service credits (10–25% of monthly bill for the affected service). Composite SLA for multi-service architectures is calculated by multiplying individual SLAs.

Azure Service	SLA Uptime	Downtime/Month (99.99%)	Credit for Breach
Azure VMs (Availability Zones)	99.99%	~4.3 minutes	10% (<99.99%) / 25% (<99%)
Azure VMs (Availability Set)	99.95%	~21.9 minutes	10% (<99.95%) / 25% (<99%)
Azure SQL Database	99.99%	~4.3 minutes	10% (<99.99%) / 25% (<99%)
Azure Cosmos DB	99.999%	~26 seconds	10% (<99.999%) / 25% (<99.99%)
Azure Blob Storage	99.9%	~43.8 minutes	10% (<99.9%) / 25% (<99%)
Azure App Service	99.95%	~21.9 minutes	10% (<99.95%) / 25% (<99%)
Azure Kubernetes Service	99.95%	~21.9 minutes	10% (<99.95%) / 25% (<99%)
Azure Active Directory (Entra ID)	99.99%	~4.3 minutes	10% (<99.99%) / 25% (<99%)
Azure Functions (Consumption)	99.95%	~21.9 minutes	10% (<99.95%) / 25% (<99%)
Azure ExpressRoute	99.95%	~21.9 minutes	10% (<99.95%) / 25% (<99%)

10.4 Azure Certification Roadmap

Microsoft Azure certifications validate your cloud skills and are organized into three levels: Fundamentals, Associate/Specialty, and Expert. Certifications are role-based and renewed every year through Microsoft Learn assessments (no re-exam required for renewal).

Exam	Certification Name	Level	Prerequisites	Key Topics
AZ-900	Azure Fundamentals	Fundamentals	None	Cloud concepts, core Azure services, security, governance, pricing, SLAs
AI-900	Azure AI Fundamentals	Fundamentals	None	AI concepts, Azure AI services, ML fundamentals, responsible AI
DP-900	Azure Data Fundamentals	Fundamentals	None	Data concepts, relational/non-relational data, Azure data services, analytics

Exam	Certification Name	Level	Prerequisites	Key Topics
AZ-104	Azure Administrator Associate	Associate	AZ-900 recommended; 6 months experience	Manage identities, governance, storage, compute, virtual networking, monitoring
AZ-204	Azure Developer Associate	Associate	AZ-900 recommended; 1-2 years dev experience	Develop Azure compute, storage, security, caching, API integration solutions
AZ-500	Azure Security Engineer Associate	Associate	AZ-900/AZ-104 recommended	Identity, networking security, compute security, data & app security
DP-203	Azure Data Engineer Associate	Associate	DP-900 recommended	Data storage, data processing, data security, monitoring data solutions
AI-102	Azure AI Engineer Associate	Associate	AI-900 recommended	Azure AI services, computer vision, NLP, speech, decision services, OpenAI
AZ-305	Azure Solutions Architect Expert	Expert	AZ-104 required	Design identity, networking, compute, storage, data, monitoring, business continuity
AZ-400	DevOps Engineer Expert	Expert	AZ-104 or AZ-204 required	DevOps practices, Agile, CI/CD, dependencies, app config, feedback loops

10.5 Quick Reference: Azure Service Selection Guide

Category	Service	Choose When...
Compute	Azure VMs	You need full OS control, running legacy apps, or lift-and-shift migration
Compute	Azure App Service	Hosting web apps or REST APIs without managing VMs; multiple runtime support needed
Compute	Azure Kubernetes Service (AKS)	Running containerized microservices at scale; GitOps/DevOps workflows
Compute	Azure Container Instances (ACI)	Running single containers on-demand without orchestration; burst workloads
Compute	Azure Functions	Event-driven, short-running code; pay-per-execution; serverless architecture
Storage	Azure Blob Storage	Unstructured data: images, videos, backups, logs, big data lake storage
Storage	Azure Files	SMB or NFS file shares; lift-and-shift file servers; shared config files for apps
Storage	Azure Queue Storage	Simple, decoupled message queuing between application components
Network	Azure Virtual Network	Private networking for Azure resources; subnetting; VNet peering
Network	Azure Application Gateway	Layer 7 load balancing; URL routing; WAF; SSL termination; multi-site hosting
Network	Azure Load Balancer (Standard)	Layer 4 TCP/UDP load balancing; zone-redundant; high-performance
Network	Azure Front Door	Global HTTP load balancing; CDN + WAF; anycast routing; geo-filtering

Category	Service	Choose When...
Network	Azure ExpressRoute	Private, dedicated, high-bandwidth connection from on-premises to Azure
Database	Azure SQL Database	Managed relational database; new cloud apps; PaaS with 99.99% SLA
Database	Azure SQL Managed Instance	Lift-and-shift SQL Server to PaaS; near-full SQL Server compatibility
Database	Azure Cosmos DB	Global distribution; multiple APIs; single-digit ms latency; flexible schema
Database	Azure Cache for Redis	In-memory caching; session state; leaderboards; pub/sub messaging
Database	Azure Synapse Analytics	Enterprise data warehouse + Big Data analytics; unified analytics platform
AI / ML	Azure OpenAI Service	Access GPT-4o, DALL-E, Whisper with enterprise security and private networking
AI / ML	Azure Machine Learning	Full MLOps lifecycle: data prep, train, register, deploy, monitor ML models
AI / ML	Azure AI Services	Pre-built AI APIs: vision, speech, language, document intelligence (no ML needed)
Security	Azure Key Vault	Secrets, encryption keys, and certificate management; eliminate hardcoded creds
Security	Microsoft Entra ID	Cloud identity: SSO, MFA, Conditional Access, B2B/B2C, app registration
Security	Microsoft Defender for Cloud	Cloud security posture management; threat protection; compliance scoring
Monitor	Azure Monitor	Unified metrics, logs, alerts, dashboards for all Azure and hybrid resources
Monitor	Application Insights	APM for web apps; distributed tracing, performance, availability monitoring
DevOps	Azure DevOps	End-to-end: boards, repos, CI/CD pipelines, artifacts, test plans
DevOps	GitHub Actions + Azure	Open-source workflow automation with native Azure deployment actions
Governance	Azure Policy	Enforce compliance rules; prevent non-compliant resource creation
Governance	Azure Cost Management	Monitor, budget, analyze, and optimize cloud spend across subscriptions

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